Lesson 1: Polynomials — Review and Preview

Terms, degree and names — then algebra tiles, like terms, and adding and subtracting polynomials.

1. Basic Vocabulary

In algebra, a letter that stands for one or more numbers is called a **variable**, and an expression such as $3m - n + 7$ or $\frac{4}{y} - 1$ is an **algebraic expression**.

A **monomial** is a number, a variable, or a product of numbers and variables. Every variable in a monomial must have a positive integer exponent, and no variable may appear in a denominator. *eg.* $7,\ y,\ 8y,\ -\frac{1}{3}pq,\ 0.75m^3,\ xyz,\ 3a^5b^2$. The number multiplying the variable(s) in a monomial is its **numerical coefficient**.

A **polynomial** is a monomial, or a sum or difference of monomials: $9,\ y,\ 9+y,\ 3m+8n,\ y^2-6y-11$. But $\frac{3}{x} - 2$ is <u>not</u> a polynomial, since it equals $3x^{-1} - 2$, and a negative exponent on a variable is not allowed.

A polynomial is made up of one or more **terms**, separated by $+$ or $-$ signs. A polynomial with 1 term is a **monomial** (eg. $5y$); with 2 terms a **binomial** (eg. $y - 6$); with 3 terms a **trinomial** (eg. $y^2 + y - 6$); and with 4 or more terms it is simply called a **polynomial**.

Example 1 — Is it a polynomial, and what is it called?

For each expression below: state whether it represents a polynomial; if it does not, explain why; and if it does, state whether it is a monomial, binomial, or trinomial.

(a) $\frac{2}{5}mn^3 + 7$ **(b)** $5x^{2/3}y$ **(c)** $\sqrt{3}\,x^5 - 2x^2 + 6$

(d) $4y^3 - 7y + 2y^{0.5}$ **(e)** $\frac{9}{b^2}$

Example 2 — Counting variables and terms

Complete the following table.

Polynomial Expression	# Variables	# Terms	Name of Polynomial
$7a + 5bc$			
$3p - 5q + 8r$			
$y^2 - 4y - 9$			
$\sqrt{5}\,m$			
$5m^3 + 2m^2n + 4n^2 - 7$			

2. Classifying Polynomials by Degree

The **degree of a monomial** is the sum of the exponents of its variable(s). *eg.* $3x^4$ has degree 4, and $-\frac{5}{6}ab^2c^3$ has degree 6.

The **degree of a polynomial** is the degree of the term with the highest degree. *eg.* $2x^2y^3 - 4x^5 + xy^4 - 8$ has degree 5. If a polynomial has a term of degree zero (no variable present), that term is called a **constant term**; in $2x^2y^3 - 4x^5 + xy^4 - 8$ the constant term is -8.

Example 3 — Building a polynomial to order

Give an example of each of the following.

(a) a binomial of degree 3 in one variable.

(b) a trinomial in two variables with a constant term.

(c) a monomial of degree 5 with a numerical coefficient of 7.

(d) a binomial of degree 10 with each term containing two variables.

3. Polynomials in a Single Variable

Polynomials in one variable are usually arranged in **ascending** order (powers increasing) or **descending** order (powers decreasing). The **leading coefficient** of a polynomial in a single variable is the numerical coefficient of the term with the highest power of the variable.

Example 4 — Ordering, leading coefficient and constant term

Consider the polynomial expression $3x - 5x^3 + 9 + \dfrac{2x^2}{3}$.

(a) Write the polynomial in descending powers of x.

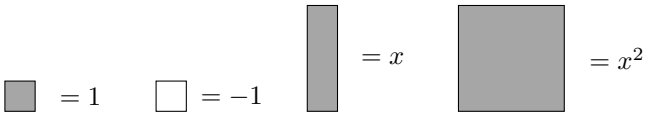
(b) Write the polynomial in ascending powers of x.

(c) State the leading coefficient and the constant term.

(d) State the numerical coefficient of the term in x^2.

4. Representing Polynomials with Algebra Tiles

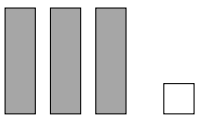
Algebra tile legend for this course: a **shaded** tile is positive, an **unshaded** (outline only) tile is negative.



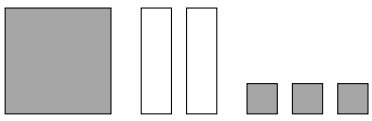
Example 5 — Reading a tile diagram

State the polynomial expression which describes each diagram below.

(a)



(b)



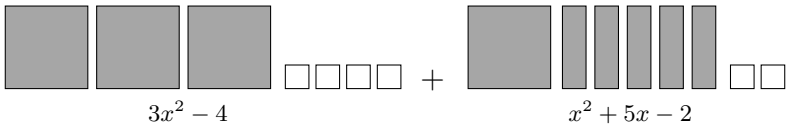
5. Adding and Subtracting Polynomials

Note: Subtracting a polynomial is the same as adding its opposite: $(5x + 2) - (3x - 7) = (5x + 2) + (-3x + 7)$.

Example 6 — Adding and subtracting with tiles

Use algebra tiles (or reasoning about zero pairs) to determine each result.

(a) Addition: $(3x^2 - 4) + (x^2 + 5x - 2)$ *The tiles for (a) are shown as a model.*



$3x^2 - 4$ $x^2 + 5x - 2$

(b) Subtraction: $(-2x^2 + 5x - 3) - (3x^2 - 2x - 4)$

Like terms are terms with the same variable(s) raised to the same exponent(s): $4p$, $9p$, p are like terms, and so are $3y^3$, $\frac{1}{4}y^3$, $-6y^3$. **Unlike terms** have different variables, or the same variable raised to different exponents: $3y^3$, $\frac{1}{4}y^2$, $-6y$ are unlike, as are $5m$ and $5n$. Only like terms may be combined into a single term.

Example 7 — Collecting like terms, horizontally and vertically

Simplify each of the following by collecting like terms.

(a) $(5p - 3q + 4r) + (2q - 5p - 7r)$

(b)
$$\begin{array}{r} 6x^2 - 5x + 4 \\ -\ (3x^2 - 8x - 2) \\ \hline \end{array}$$

Example 8 — Collecting like terms in longer expressions

Simplify.

(a) $7y - 3y^2 + 4 - 5y^2 + 9 - 2y$

(b) $m^2n - mn^2 + 5m^3n - 6mn^2 + 4m^2n$

Lesson 2: Multiplying Monomials and Binomials

Area models and the distributive property — from $a(b + c)$ all the way to FOIL.

1. The Area Model

The area of a rectangle is (length)×(width), and cutting a rectangle into two side-by-side pieces splits its area into the sum of the two smaller areas. That gives a picture for multiplying a monomial by a binomial: build a rectangle with those two side lengths, split it along the binomial's two terms, and add the pieces.

	x	2
5	$5x$	10

The big rectangle has width $x + 2$ and height 5, so its area is $5(x + 2)$; the pieces of width x and 2 have areas $5x$ and 10. Both describe the same rectangle, so $5(x + 2) = 5x + 10$.

Negative entries. When a factor contains a subtraction, as in $-2(x - 4)$ in Example 1(c), we keep the same picture and write the signed entries (-2 and -4) on its sides. A real length can't be negative, so read the picture as a **bookkeeping** area model: it keeps track of every product, and a negative sign on an entry simply gives that piece a negative area.

Example 1 — Monomial times binomial, drawn as an area

Use an area model to determine each product. Sketch the rectangle, label the two side lengths of each piece, and state the resulting sum (or difference) of areas.

(a) $4(x + 3)$ **(b)** $x(x + 5)$ **(c)** $-2(x - 4)$

(d) $3x(x - 1)$ **(e)** $(x - 3)(2x)$

2. The Distributive Property

Each product in Example 1 — $4(x + 3) = 4x + 12$, $x(x + 5) = x^2 + 5x$, $-2(x - 4) = -2x + 8$, $3x(x - 1) = 3x^2 - 3x$, $(x - 3)(2x) = 2x^2 - 6x$ — is an instance of the **distributive property**, $a(b + c) = ab + ac$, which extends to any number of terms and may also be applied with the monomial written second: $(b + c)a = ba + ca$.

Example 2 — Distributing over two and three terms

Use the distributive property to determine the following products.

(a) $5(4x + 2)$ **(b)** $-3(3x^2 + x - 5)$

(c) $(x^3 - 3)x^2$ **(d)** $-4x(5x - 3y + z)$

Writing a *product* of polynomials as a *sum or difference of terms*, as in each part above, is called **expanding**: it combines the distributive property $a(b + c) = ab + ac$ with the exponent rule $x^a \times x^b = x^{a+b}$.

3. Expanding and Simplifying

When an expression contains more than one set of brackets, expand each bracket first, then combine any resulting like terms.

Example 3 — Expand every bracket, then collect

Expand and simplify.

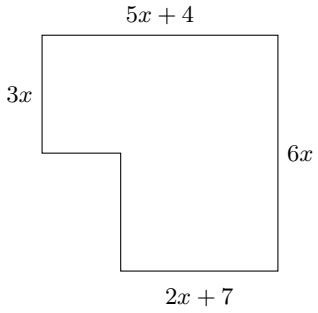
(a) $8 - 3(5x + 2)$ **(b)** $3(4x - 2) - 3(x - 5)$ **(c)** $4x(2x^2 - 5x + 3) - (6x + 2x^2)$

4. Applying the Distributive Property: Area of Composite Shapes

The distributive property gives a simplified area expression for an irregular (composite) shape two ways: by **adding** the areas of two rectangles that make up the shape, or by **subtracting** a missing corner from a larger rectangle. Both must give the same expression.

Example 4 — One shape, two methods

Determine a simplified expression for the area of the shape below.

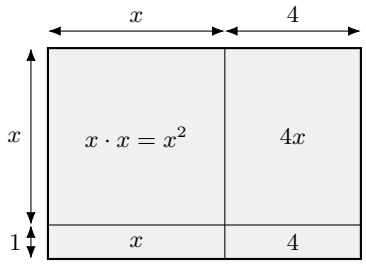


(i) by adding the areas of two rectangles

(ii) by subtracting the area of one rectangle from another

5. Multiplying Binomials Using an Area Model

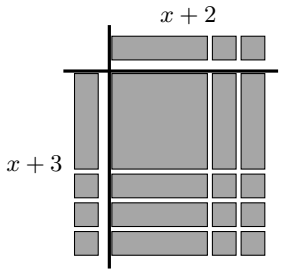
Think of $(x+4)(x+1)$ as the area of a rectangle of length $x+4$ and width $x+1$. Splitting the length into x and 4 and the width into x and 1 breaks the rectangle into four smaller ones — an x-by-x square, an x-by-4 strip, an x-by-1 strip and a 4-by-1 block — and adding their areas gives the product in expanded form.



×	x	4
x	x^2	$4x$
1	x	4

$(x+4)(x+1) = x^2+4x+x+4 = x^2+5x+4$. The rectangle does not have to be drawn to scale every time: the **four-cell grid** on the right is a quick sketch of the same **area model**, with the pieces of one factor across the top and the pieces of the other down the side. In this course both pictures are called an area model.

Multiplying with algebra tiles. Algebra tiles build the same area model out of real pieces (Lesson 1's legend: shaded is positive, unshaded is negative). Lay the tiles for one factor across the top of a frame and the tiles for the other factor down the side. Then fill the rectangle: each tile inside is the product of the factor tile above it and the factor tile beside it ($x \cdot x$ gives an x^2-tile, $x \cdot 1$ an x-tile, $1 \cdot 1$ a unit tile). **Shading rule:** two factor tiles with the *same* shading give a shaded (positive) tile; *different* shading gives an unshaded (negative) tile. Count the tiles inside the frame to read the product, removing any zero pairs.



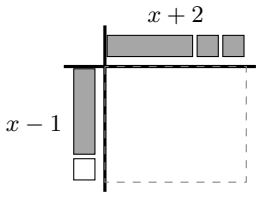
Top: $x+2$ (one x-tile, two unit tiles)
Side: $x+3$ (one x-tile, three unit tiles)
Inside: one x^2-tile, $2+3=5$ x-tiles and $2\times 3=6$ unit tiles, so

$$(x+2)(x+3) = x^2+5x+6.$$

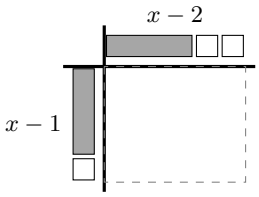
Example 5 — Multiplying with algebra tiles

The factor tiles are already in place. Fill in each rectangle using the shading rule, then state the product. Look for zero pairs before you write the answer.

(a) $(x+2)(x-1)$



(b) $(x-2)(x-1)$



Example 6 — Area models as four-cell grids

Use an area model to determine each product.

(a) $(x+5)(x+2)$ **(b)** $(a^2-3)(a^2-7)$ **(c)** $(2p+3q)(p+8q)$

The same grid multiplies two-digit whole numbers without a calculator: split each number into tens and ones, then add the four partial products.

×	40	1
50	2000	50
3	120	3

$41 \times 53 = 2000 + 120 + 50 + 3 = 2173$

Why this works. Let $t = 10$. Then $41 = 4t + 1$ and $53 = 5t + 3$, so

$$41 \times 53 = (4t+1)(5t+3) = 20t^2 + 17t + 3 = 2000 + 170 + 3 = 2173.$$

Multiplying two-digit numbers *is* multiplying two binomials, with $t = 10$ in place of x.

Example 7 — The same grid on two-digit numbers

Use an area model and no calculator to determine each product.

(a) 32×45 **(b)** 67×84

6. Multiplying Binomials Using the Distributive Property and FOIL

Distributing a whole binomial works exactly like distributing a monomial:

$$(a+b)(c+d) = a(c+d) + b(c+d) = ac + ad + bc + bd.$$

These are the same four terms the area model produced: the grid and the distributive property just organize the same calculation differently.

Example 8 — Distributing a binomial

Use the distributive property to determine each product.

(a) $(x+5)(x+3)$ **(b)** $(a-6)(3a-2)$

(c) $(m-9)(n-9)$ **(d)** $(x+3y)(x-8y)$

Because those four products always follow the same pattern, they can be remembered with the acronym **FOIL**: **F**irst terms (ac), **O**utside terms (ad), **I**nside terms (bc), **L**ast terms (bd). FOIL is not a new method — it is a memory device for applying the distributive property to two binomials in a fixed order, so no term is missed. FOIL only works for a *binomial times a binomial* (exactly $2 \times 2 = 4$ products). A binomial times a trinomial (Lesson 4) has six products, and FOIL would miss two of them.

Example 9 — FOIL

Use FOIL to determine each product.

(a) $(x+7)(x+2)$ **(b)** $(y-4)(y+6)$

(c) $(2x+3)(x-5)$ **(d)** $(4a-3b)^2$

Tip — check by substitution. Substitute a small number such as $x = 2$ into both the product and your expansion; they must give the same value. For Example 9(a): $(2+7)(2+2) = 36$ and $2^2 + 9(2) + 14 = 36$. ✓

Example 10 — Three methods, one answer

A rectangle has length $(3a+2)$ cm and width $(a+5)$ cm. Determine the area of the rectangle, in cm^2, three ways, and confirm that all three methods agree.

(i) use an area model **(ii)** use the distributive property **(iii)** use FOIL

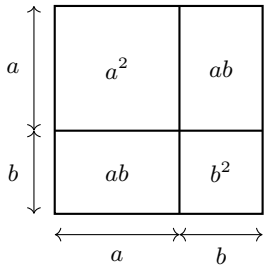
Check: all three methods should give the same expression.

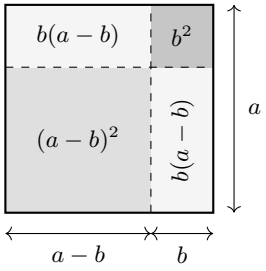
Lesson 3: Multiplying Polynomials I — Special Products

Squaring a binomial, sum times difference — derived, recognized on sight, then used in reverse.

1. Two Patterns Worth Memorizing

Picture a square of side length $a+b$ (left picture). Its area is $(a+b)^2$, but the same square also slices into four pieces — two squares and two equal rectangles — whose areas we can add instead: $a^2+ab+ab+b^2 = a^2+2ab+b^2$, so $(a+b)^2 = a^2+2ab+b^2$.





For $(a-b)^2$ (right picture), start with a square of side a and cut a strip of width b off the right side and another off the top. Each strip has area ab, but the two strips overlap in the b-by-b corner, so that corner has been cut away twice; add it back once: $(a-b)^2 = a^2 - ab - ab + b^2 = a^2 - 2ab + b^2$. Multiplying a sum by a difference of the *same* two terms makes the two middle terms of FOIL cancel: $(a-b)(a+b) = a^2 + ab - ab - b^2 = a^2 - b^2$.

Watch out: $(a+b)^2 \neq a^2 + b^2$**.** Try numbers: $(3+4)^2 = 7^2 = 49$, but $3^2 + 4^2 = 9 + 16 = 25$. The missing 24 is the two ab rectangles in the left picture ($2 \times 3 \times 4 = 24$). Squaring a binomial always produces a middle term.

Two shortcuts to remember.

- **Squaring a binomial:** square the first term, double the product of the two terms, then square the last term.

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a-b)^2 = a^2 - 2ab + b^2$$

- **Sum times difference:** multiplying the sum and the difference of the same two terms always gives the difference of their squares.

$$(a+b)(a-b) = a^2 - b^2$$

Example 1 — Recognizing the two patterns

For each product, first decide whether one of the two patterns applies. If it does, use it to expand; if not, expand with FOIL.

(a) $(x+9)^2$ **(b)** $(4x-1)^2$ **(c)** $(3m-2n)^2$

(d) $(6-5x)(6+5x)$ **(e)** $(4a+7b)(4a-7b)$ **(f)** $(3p-8q)(3p-8q)$

(g) $(x+3)(x-5)$ **(h)** $(-2x+5)^2$

2. A Coefficient Out Front

A special product may be multiplied by a number (or a negative sign) sitting outside the brackets. Expand the binomial product *first*, then distribute the outside factor across every resulting term.

Example 2 — Expand first, then distribute

Expand and simplify.

(a) $3(4x-5)(x-2)$ **(b)** $-6(5p+2)^2$

3. Combining Two or Three Products

A longer expression may add or subtract several products in a row. Expand *every* product completely — a subtraction sign in front of a product flips the sign of every term that comes out of it — and only then collect like terms.

Example 3 — Several products at once

Expand and simplify.

(a) $(x+6)(x-6)-(x+3)(x+9)$

(b) $4x(2x^2-5x+3)-(2x^2+3x)(4x-7)$

(c) $3(3x-4)(2x+3)-5(x-2)(4x-3)$

4. Extension — Matching the Form $(x+a)^2+b$

(Extension: rewriting a trinomial this way is called completing the square, a Math 20-1 skill.) Because $(x+a)^2=x^2+2ax+a^2$, any trinomial x^2+kx+c can be rewritten as $(x+a)^2+b$: choose a so that $2a$ matches the middle coefficient k, then adjust b so the two expressions agree once expanded.

Example 4 — Extension: using the square pattern in reverse

For every value of x, the polynomial $x^2+16x+70$ can be written in the form $(x+a)^2+b$. Determine the values of a and b.

Lesson 4: Multiplying Polynomials II — Binomial × Trinomial and Three Binomials

Six partial products at a time — by grid, by distribution, and three factors at once for volume.

1. Multiplying a Binomial by a Trinomial

When a binomial (2 terms) is multiplied by a trinomial (3 terms), every term of the binomial must "shake hands" with every term of the trinomial: $2\times3=6$ partial products before any like terms are collected. A small grid keeps track of all six — one factor labels the rows, the other labels the columns, and each cell holds the product of its row and column labels. Adding every cell and combining like terms gives the expansion.

> **General rule for multiplying polynomials.** Multiply *every* term of the first factor by *every* term of the second, then collect like terms. An m-term polynomial times an n-term polynomial gives $m\times n$ partial products before collecting: binomial × binomial: $2\times2=4$ (this is FOIL); binomial × trinomial: $2\times3=6$; trinomial × trinomial: $3\times3=9$. FOIL is only the 2×2 case.

Example 1 — The same product two ways: grid, then distribution

A rectangle has length $(3x+4)$ cm and width $\left(x^2+3x+2\right)$ cm. Determine the area of the rectangle, in cm^2, using

(i) an area model (grid)

(ii) the distributive property.

Example 2 — Expanding a binomial times a trinomial

Expand and simplify.

(a) $\left(x^2 - 9\right)\left(3x^3 + 2x - 7\right)$

(b) $\left(3y^2 - 2y - 5\right)(y - 4)$

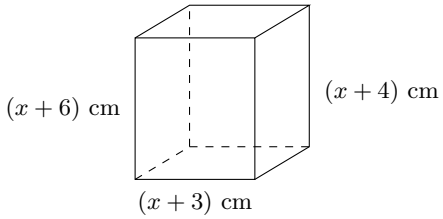
(c) $3(b - 5)\left(b^2 - 2b - 4\right) - (5b - 3)(5b + 3)$

2. Multiplying Three Binomials

The same idea extends to three binomials: multiply the first two, multiply that result by the third, then collect like terms. Because multiplication is commutative and associative, neither the choice of which pair to multiply first nor the order of the factors changes the final expansion. This is how we find the **volume** of a rectangular prism whose dimensions are binomials, since $V = \text{length} \times \text{width} \times \text{height}$.

Example 3 — Volume of a rectangular prism

Consider the rectangular prism shown, with edge lengths $(x + 3)$ cm, $(x + 4)$ cm, and $(x + 6)$ cm.



(a) Write an expression for the volume of the prism, in cm^3.

(b) Expand the expression in (a) to write the volume in the form $V = ax^3 + bx^2 + cx + d$.

Example 4 — The order of the three factors does not matter

Expand and simplify.

(a) $(x-2)(x+5)(3x-1)$

(b) $(3x-1)(x-2)(x+5)$ (multiply $(3x-1)(x-2)$ first)

(c) Comment on the results to (a) and (b).

Lesson 5: Applying Polynomial Products to Area, Volume, and Equations

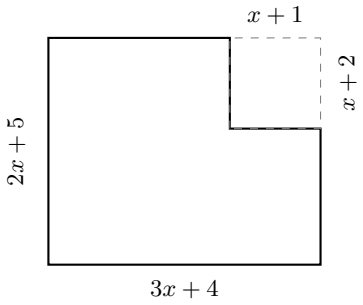
Composite areas, framed regions, and geometric equations that must be expanded before they can be solved.

1. Composite Area Figures

A figure made entirely of horizontal and vertical segments has two common strategies for writing its area as a single polynomial. **Strategy A — add the pieces.** Slice the figure into non-overlapping rectangles, find each area as a product of its side lengths, and add. **Strategy B — subtract from a bounding box.** Take the smallest rectangle that contains the figure and subtract the missing corner piece (the "notch"). Both give the same polynomial once expanded; choose whichever is easier to see in a given figure.

Example 1 — Bounding box minus a notch

The L-shaped figure below is the piece left over after a rectangular notch is cut from the top-right corner of a larger rectangle.



(a) Write the area of the figure as a trinomial ax^2+bx+c.

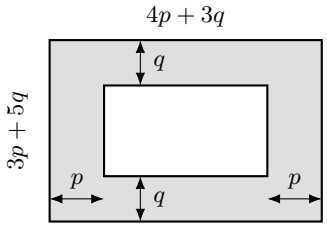
(b) Evaluate the area when $x=2$.

2. Framed Regions

A rectangular "frame" is a rectangle with a smaller rectangle removed from its interior; the shaded (remaining) area is the outer area minus the inner area.

Example 2 — A frame with two different widths

In the figure below, a rectangular frame surrounds a smaller rectangular opening. The frame's sides are p wide and its top and bottom are q wide, so the opening is set in by p on the left and right and by q along the top and bottom.



(a) Determine a simplified expression, in terms of p and q, for the shaded area.

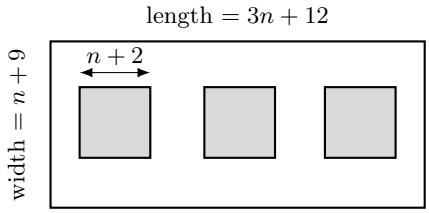
(b) Calculate the shaded area when $p = 2$ and $q = 1$.

3. Modelling with Squares and Rectangles

Many real-world area problems are modelled the same way: a large region with one or more smaller regions removed from inside it.

Example 3 — Garden with three flower beds

A rectangular garden has length $(3n + 12)$ m and width $(n + 9)$ m. Three square flower beds, each with side length $(n + 2)$ m, are planted inside the garden. The rest of the garden is grass.



(a) Write and simplify an expression for the area of grass in the garden.

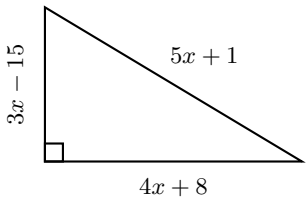
(b) Determine the area of grass when $n = 2$, and check your answer using the dimensions.

4. Equation Solving

Some problems describe a figure with polynomial expressions plus a geometric *rule* that must hold — for example the Pythagorean theorem for a right triangle. Expanding both sides of the resulting equation often cancels the highest-degree terms, leaving a linear equation to solve.

Example 4 — The Pythagorean theorem with polynomial sides

The hypotenuse of a right triangle is $(5x + 1)$ cm long, and the lengths of the other two sides are $(4x + 8)$ cm and $(3x - 15)$ cm.



Form an equation using the Pythagorean theorem, and solve it to determine the lengths of all three sides of the triangle.

Lesson 6: Polynomial Operations Unit Review

Every skill in the unit, one worked example at a time — classify, expand, apply, and catch the classic errors.

1. Classifying Polynomials and Finding Degree

Recall: a **monomial** has exactly one term, a **binomial** two, and a **trinomial** three. Every monomial, binomial and trinomial is a **polynomial**; with 4 or more terms we just say "polynomial". The **degree of a monomial** is the sum of the exponents on its variables, and the degree of a polynomial is the largest degree among its terms. A variable with a *negative* exponent (like x^{-2}) is not allowed in a polynomial term, so an expression containing one is not a polynomial.

Example 1 — Monomial or not, and finding degree

Determine each of the following.

(a) Which of the following could *not* be classified as a monomial?
A. $8x^3$ B. $x-5$ C. -3 D. $9y$

(b) Find the degree of the polynomial $m^4n^2+5p^3$.

(c) Which of the following is a polynomial of degree 6?

I. $7x^2$ II. $4x^6-3x^8$ III. $2x^3-5x^{-2}+6x^4$ IV. $4x^2y^4$

A. IV only B. III and IV only C. I, II, III, and IV D. some other combination

Example 2 — Leading coefficient, degree and constant term

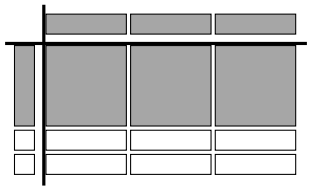
The polynomial $4x^3-15+6x^4$ has leading coefficient P, degree Q, and constant term R. Find the value of $P-Q-R$.

2. Multiplying with Algebra Tiles

An algebra-tile model pictures a distributive-property multiplication (Lesson 2, Section 5): the factor tiles go across the top and down the side of a frame, and the product fills the rectangle. An x^2-tile is x^2, an x-tile is x, a unit tile is 1, and the shading gives the sign — **shaded** positive, **unshaded** negative. Reading the picture as "rows times columns" recovers the product.

Example 3 — Reading a tile array as a product

The algebra-tile model below shows the factor tiles across the top and down the side, and the product filled in: three shaded x^2-tiles and six unshaded x-tiles.



The diagram represents the expansion of A. $3x^2(x-2)$ B. $x^2(x-2)$ C. $3x(x-2)$ D. $3x(x+2)$

3. Multiplying Binomials

Multiplying two binomials produces up to four terms; when the "outer" and "inner" products are opposites, as in $(x+y)(x-y)$, they cancel and leave a two-term **difference of squares**. To check an expansion, verify each of the four partial products separately: exponents add, signs multiply.

Example 4 — Counting terms, and auditing an expansion

(a) How many terms remain after fully expanding and simplifying $(2x-5)(2x+5)$?

(b) Exactly one of the following expansions is correct. Which one?

A. $-4x^2(2x^3+3) = -8x^6 - 12x^2$
B. $-6x(3-2x) = -18x - 12x^2$
C. $-3x^3(5+x) = -15x^3 - 3x^3$
D. $-8x^2(1-x) = -8x^2 + 8x^3$

4. Applying Multiplication to Area

Example 5 — Area of a rectangle with binomial sides

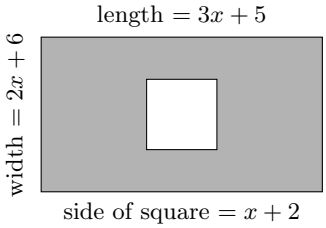
The area (in mm^2) of a rectangle with length $5-b$ mm and width $9-b$ mm is
A. $45+b^2$ B. $45-4b-b^2$ C. $45-4b+b^2$ D. $45-14b+b^2$

Example 6 — Area of an isosceles right triangle

Triangle PQR is isosceles and right-angled at Q, with legs $PQ = QR = 2x+6$ cm. If the area of the triangle can be written as ax^2+bx+c cm^2, find the value of b. A. 24 B. 12 C. 6 D. 4

Example 7 — A rectangle with a square cut out

Use the following diagram. A rectangle has length $3x+5$ mm and width $2x+6$ mm. A square of side length $x+2$ mm has been cut out of its interior, leaving the shaded region shown.



The area, in mm^2, of the shaded region is
A. $5x^2+28x+30$
B. $5x^2+32x+26$
C. $5x^2+24x+26$
D. $5x^2+24x+34$

5. The Distributive Property with Several Terms

Expand each group separately first, then combine like terms at the very end. Watch signs, especially when subtracting a group enclosed in brackets.

Example 8 — Three groups, then collect

The expression $5x(3-2x)-4x(x-3)-(7x+5)$ can be written in the form ax^2+bx+c. Find the value of abc.

6. Multiplying a Binomial by a Trinomial, and Combining Results

Example 9 — Binomial times trinomial, and a sum minus a product

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(a) Expand $(5t-2)(3t^2-4t-2)$.

A. $15t^3-26t^2-2t+4$

B. $15t^3-34t^2+2t+4$

C. $15t^3-26t^2+2t+4$

D. $15t^3-26t^2-2t-4$

(b) Subtracting the product of $(3x-2)$ and $(2x+3)$ from the sum of $(4x^2+2x-3)$ and $(2x^2+4x-5)$ results in a polynomial of the form px^2+qx+r. How many of the parameters p, q, r are equal to zero? A. 0 B. 1 C. 2 D. 3

7. Multiplying Three Binomials

To expand a product of three binomials, multiply any two together first, then multiply that result by the third factor.

Example 10 — Three factors, two stages

Expand $(a+2)(a-5)(3a+1)$.

A. $3a^3-8a^2-33a-10$

B. $3a^3+10a^2-27a-10$

C. $3a^3+a^2-39a-10$

D. $3a^3-10$

8. Extension — Completing the Square

(Extension: completing the square is a Math 20-1 skill and is not part of the Math 10C review.) Every trinomial x^2+bx+c can be rewritten as $(x-p)^2+q$: take p to be half the coefficient of x with its sign flipped, then adjust the constant to match.

Example 11 — Extension: matching coefficients, and completing the square

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Use the square-of-a-binomial pattern in reverse.

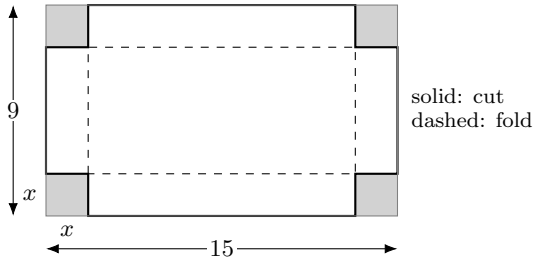
(a) If $(x-a)^2+a=x^2-10x+c$, find the value of c.

(b) The trinomial $x^2-16x+70$ can be expressed in the form $(x-p)^2+q$. Find the values of p and q.

9. An Application: Volume of a Folded Box

Example 12 — Cut the corners, fold up the sides

Use the following information. A sheet of paper measures 15 cm by 9 cm. Squares of side x cm (shaded) are cut from each corner, as shown, and the paper is folded along the dashed lines to form an open rectangular box.



(a) The length and width (in cm) of the box are, respectively,

A. $15 - x$ and $9 - x$

B. $15 - 2x$ and $9 - 2x$

C. $15 + x$ and $9 + x$

D. $15 + 2x$ and $9 + 2x$

(b) The volume of the box can be written as the polynomial $ax^3 + bx^2 + cx$, where a, b, and c are integers. Find the value of $a + c$.

10. Common Errors When Expanding Squared Binomials

A very common mistake is distributing a number *into* a squared binomial before squaring — order of operations applies the square first. Others are squaring a binomial by doubling its first coefficient instead of squaring it, and forgetting the factor of 2 in the middle term of $(mx - n)^2 = m^2x^2 - 2mnx + n^2$.

Example 13 — Three students, three different errors

Priya, Marcus, and Devon each try to expand $6(3y - 2)^2$. Their work is shown below.

Priya	**Marcus**	**Devon**
$6(3y - 2)^2$	$6(3y - 2)^2$	$6(3y - 2)^2$
$(18y - 12)^2$	$6(6y^2 - 12y + 4)$	$6(9y^2 - 6y + 4)$
$324y^2 - 432y + 144$	$36y^2 - 72y + 24$	$54y^2 - 36y + 24$

Each student's work contains exactly one mathematical error.

(a) Describe in detail the error Priya made.

(b) Describe in detail the error Marcus made.

(c) Describe in detail the error Devon made.

(d) Determine the correct expansion of $6(3y - 2)^2$.

(e) Verify your expansion by substituting $y = -3$ into both the original expression and your expansion, confirming the same numerical value.